

ISC Class XII Maths — Chapter 3

Matrices — Formula Sheet

A. Order, Types & Counting

Matrix of order $m \times n$ has mn elements, general element a_{ij} (i = row, j = column).

Row matrix: $1 \times n$ **Column matrix:** $m \times 1$ **Square matrix:** $m = n$

Diagonal: square, $a_{ij} = 0$ for $i \neq j$ **Scalar:** diagonal with all a_{ii} equal **Identity I :** diagonal, all $a_{ii} = 1$

Total number of $m \times n$ matrices possible with each entry chosen from k given values = k^{mn} .

B. Equality of Matrices

$A = [a_{ij}]$ and $B = [b_{ij}]$ are equal $\iff$ same order and $a_{ij} = b_{ij}$ for every i, j .

Tip: Equate corresponding entries one pair at a time — this turns a matrix equation into a system of ordinary algebraic equations you already know how to solve.

C. Addition, Subtraction & Scalar Multiplication

$A \pm B = [a_{ij} \pm b_{ij}]$ (only if A, B have the *same order*)
 $kA = [k a_{ij}]$
 $kA + mB = [k a_{ij} + m b_{ij}]$

Tip: Addition is commutative and associative: $A + B = B + A$, $(A + B) + C = A + (B + C)$. Use this freely to regroup terms when isolating an unknown matrix.

D. Multiplication — Compatibility & Entry Formula

$P_{m \times n} \cdot Q_{n \times p} = R_{m \times p}$ — defined **only if** inner dimensions match ($n = n$); result takes the outer dimensions.

$(AB)_{ij} = \sum_k a_{ik} b_{kj} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B)$

E. Properties of Matrix Multiplication

Associative: $(AB)C = A(BC)$ **Distributive:** $A(B + C) = AB + AC$

Identity: $AI = IA = A$ **Generally NOT commutative:** $AB \neq BA$

Tip: Never cancel a common matrix factor or assume $AB = BA$ unless it is proved for that specific pair. $AB = O$ does *not* imply $A = O$ or $B = O$.

Worked Pattern: Reducing Matrix Polynomials

Given a relation like $A^2 = 4A - 7I$ (from $A^2 - 4A + 7I = O$):

1. Write the higher power as a product: $A^3 = A \cdot A^2$.
2. Substitute the given relation for A^2 , expand.
3. Reduce again if an A^2 term reappears, until only A and I remain.
4. Substitute the actual matrix entries for A and I only at the *final* step.

Worked Pattern: Mathematical Induction on Matrices

To prove $P(n) : A^n = (\text{formula in } n)$ for all $n \in \mathbb{N}$:

1. **Base case** — verify $P(1)$: check A^1 equals the formula at $n = 1$.
2. **Inductive hypothesis** — assume $P(k)$ true: $A^k = (\text{formula in } k)$.
3. **Inductive step** — write $A^{k+1} = A^k \cdot A$, substitute the hypothesis, multiply out, and show it equals the formula at $n = k + 1$.
4. Conclude by the Principle of Mathematical Induction.

F. Solving Matrix Equations

Linear combinations (e.g. $2X + Y = B$, $X + Y = A$, $X - Y = B$): treat X, Y as whole matrices and eliminate/substitute exactly as with simultaneous linear equations, *then* solve entry-by-entry if needed.

Multiplicative (e.g. find B given $BA = I$ or $B(\cdot) = kI$): let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, multiply out, then equate entries to get simultaneous equations in a, b, c, d .

G. Transpose Properties

$(A')' = A$ $(kA)' = kA'$ $(A \pm B)' = A' \pm B'$

Reversal law: $(AB)' = B'A'$ $(ABC)' = C'B'A'$

Tip: The reversal law flips *both* the order of multiplication and which factor is transposed — the single most common slip is writing $(AB)' = A'B'$ instead of $B'A'$.

H. Symmetric & Skew-Symmetric Matrices

Symmetric: $A' = A \iff a_{ij} = a_{ji}$ for all i, j (mirror about the leading diagonal).
Skew-symmetric: $A' = -A \iff a_{ij} = -a_{ji}$; in particular $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$ — every diagonal entry is 0.

Worked Pattern: Symmetric + Skew Decomposition

Every square matrix A splits uniquely as $A = P + Q$ where $P = \frac{1}{2}(A + A')$ is **symmetric** (check: $P' = \frac{1}{2}(A' + A) = P$)
 $Q = \frac{1}{2}(A - A')$ is **skew-symmetric** (check: $Q' = \frac{1}{2}(A' - A) = -Q$)
Verify $P + Q = A$ as the final check.

I. Orthogonal & Rotation Matrices

A is **orthogonal** if $A'A = AA' = I$.
Rotation form $M(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ satisfies $M(x)M(y) = M(x + y)$.
Standard identity used throughout: $\sin^2 \theta + \cos^2 \theta = 1$, and $\cos A \cos B \mp \sin A \sin B = \cos(A \mp B)$, $\sin A \cos B \pm \cos A \sin B = \sin(A \pm B)$.

J. Common Traps to Avoid

- Check order compatibility *before* multiplying — AB can exist while BA does not.
- $(AB)' = B'A'$, never $A'B'$ — the reversal law reverses order.
- $A^2 = A \cdot A$ using *matrix* multiplication, never entrywise squaring.
- In induction, the inductive step must start from $A^{k+1} = A^k \cdot A$ (or $A \cdot A^k$) — don't jump straight to the claimed formula.
- A skew-symmetric matrix always has zero diagonal — use this as a quick sanity check.

K. Quick Recognition Guide

Pattern seen	Use
Matrix built from $a_{ij} = f(i, j)$	Sec. A — compute entry by entry
Two matrices set equal	Sec. B — equate entries
Sum/difference/ kA of matrices	Sec. C
Product of two matrices, or find one entry of it	Sec. D entry formula
Prove $(AB)C = A(BC)$ or $A(B+C) = AB+AC$	Sec. E properties
Given $A^2 =$ linear in A, I ; find $A^3, A^4, \dots$	Worked Pattern: Polynomial Reduction
“Prove $A^n = \dots$ for all n ”	Worked Pattern: Induction
Find unknown matrix X from linear equation(s)	Sec. F linear combination
Find B given $BA = I$ or similar	Sec. F multiplicative, assume entries
Any question with $(\cdot)'$ transpose in it	Sec. G
“Show matrix is symmetric/skew-symmetric”	Sec. H definition
“Express A as sum of symmetric + skew”	Worked Pattern: Decomposition
“Verify $A'A = I$ ” or rotation matrix $M(\theta)$	Sec. I orthogonal/rotation